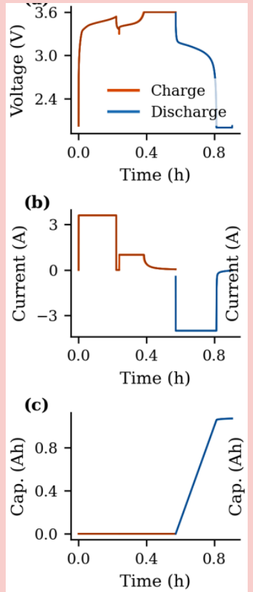
1. Input & Preprocessing

Raw Half-Cycle Data
Irregular sampling

Resample
fixed length $T = 300$

Multi-Channel Representation
(3×300): V, dv/dt , Q_{cum}

Cycle Sequence Construction
Sliding window of L cycles
$X \in \mathbb{R}^{L \times 3 \times 300}$



2. TinyBatteryNet Architecture

Multi-Scale DS-Conv Pyramid

Depthwise Conv (k = 15) → Pointwise Conv (1×1) → BN + ReLU
Depthwise Conv (k = 31) → Pointwise Conv (1×1) → BN + ReLU
Depthwise Conv (k = 61) → Pointwise Conv (1×1) → BN + ReLU

Concat C = 96

Different receptive fields capture short-, mid-, and long-term waveform patterns at multiple time scales.

SE Channel Attention

Global Average Pooling → FC (C/r) → ReLU → FC (C) → Sigmoid

Channel Weights (C)

Re-weights informative channels adaptively across chemistry domains.

Learnabe Cycle gate

GRU (96) → Linear → Sigmoid

Gate (g_t)

Suppresses early unreliable cycles (e.g., activation phase) and focuses on informative aging cycles.

Temporal Modeling

GRU (96) → Layer Norm

Models long-term degradation trajectories.

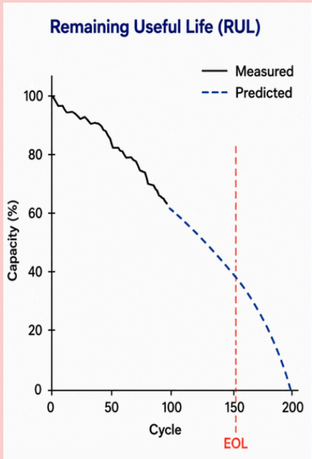
3. Prediction & Learning

RUL Prediction Head

FC (64) → ReLU → Dropout (0.3) → FC (1) → ŷ (Predicted RUL)

Output

Remaining Useful Life (cycles)



Training Loss

KDE-Weighted MAPE (Main Loss)
+
Uniform MAPE (Regularization)